

SUANR v1: Stochastic Uncertainty-Aware Neural Regression

Abstract

This document presents SUANR v1, a stochastic uncertainty-aware regression framework designed to enhance predictive modeling by incorporating uncertainty estimation alongside point predictions. The model is evaluated on a standard regression benchmark and compared against classical machine learning baselines.

Method

SUANR v1 introduces stochastic mechanisms to estimate predictive uncertainty in regression tasks. The model produces both a point prediction and an associated uncertainty signal derived from internal stochastic variation.

Experimental Setup

Experiments are conducted on a diabetes regression dataset using a repeated evaluation protocol (100 runs). Models compared include Linear Regression, Random Forest, and SUANR v1. Performance is measured using MAE and RMSE.

Results

Model	MAE	RMSE
Linear Regression	44.5463	54.8911
Random Forest	47.0133	57.9552
SUANR v1	46.9497	57.9000

Discussion

SUANR v1 demonstrates performance comparable to Random Forest but does not yet outperform Linear Regression on error metrics. However, it introduces explicit uncertainty estimation, which provides additional predictive information not available in standard baselines. This establishes SUANR v1 as a valid foundation for further development (e.g., adaptive uncertainty modeling in SUANR v2).

Conclusion

SUANR v1 serves as a validated baseline for uncertainty-aware regression. While predictive accuracy is comparable to classical models, its main contribution lies in integrating stochastic uncertainty estimation into the regression pipeline.